

Can models fitted to small data samples lead to maintenance policies with near-optimum cost?

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In a general context of maintenance modelling, the excess cost of operating at the estimated optimum value of some decision variable (as opposed to the true unknown optimum) is related to the sample size of the dataset available for modelling. Although large sample sizes are required to estimate the optimum value of a decision variable to a high accuracy, it is shown that the cost which is optimised can be reduced to near its true minimum value even for small sample sizes. In fact, the expectation of the excess cost is inversely proportional to sample size. This implies that even modest sample sizes would be sufficient for the practical use of sensible models for determining cost-based maintenance policies. An example relating to a simple model for determining the optimum inspection interval for a repairable system illustrates these ideas. In particular, some results on the maintenance of medical equipment are presented.

1. Introduction

Mathematical modelling can be used to consider the benefits of inspecting and/or replacing components or systems (of equipment), and hence to derive the optimal value of some decision variable defining, for example, inspection policy. Such a decision variable would in general be a vector quantity. Model parameters may be estimated from records of failure times and number of defects found at inspections and/or failures, for example. Such data samples are collected for a system which has been operated for some time under some (usually suboptimal) maintenance policy. The quantity of objective data available for modelling we call the *sample size*. In general, the sample size required to estimate the optimum value of a decision-variable vector to a given accuracy may be determined by considering the variance of the estimate. More importantly, the cost implications of operating at the estimated optimum value of some decision variable vector, as opposed to the true unknown optimum, can be considered for a given sample size. This is done by determining the upward bias of the cost estimate. This bias can be interpreted as the expected excess cost of operating the inspection policy, for example, at the estimated optimum, over the cost of operating at the true unknown optimum. An increasing sample size would result, under regular conditions, in a decrease in the bias. In this way the sample size may be related to the excess cost of sub-optimal operation. In the context of decision theory, this excess cost is an example of a loss function, and its expectation is sometimes called the risk. See for example DeGroot (1970 : p. 137).

It is assumed throughout that the modeller uses some statistically orthodox method of estimating model parameters from data, such as maximum likelihood. All that is

required is that the cost can be expressed as a function of decision variables and model parameters (and that this function is differentiable at least numerically to second order), and that the covariance matrix of model parameters can be estimated. These conditions are described fully in the next section.

Throughout we consider the cost associated with some maintenance policy. However, the results apply equally when the focus is on downtime; in fact, cost and downtime may be used interchangeably here.

The main results are presented in a general context in the next section. We then give an example of excess cost-calculation for a simple but relevant 'delay-time' model which is used to determine the optimum inspection interval for a repairable system. Finally, a numerical example of the application of this model to medical equipment is presented. The sample size was typical of that occurring in practical studies.

2. The general case

In general we seek to minimize a cost (or downtime) measure D , where $D = D(\delta_1, \dots, \delta_p; \theta_1, \dots, \theta_q)$, that is D is a function of p decision variables and q model parameters. For example, $\delta = (\delta_1, \dots, \delta_p)$ may be the vector of times of the next p inspections. The model parameter vector $\theta = (\theta_1, \dots, \theta_q)$ would require estimation through the fitting of some suitable model. As the model parameters are not themselves of direct interest, we shall refer to them as nuisance parameters. In vector notation, we have $D = D(\delta; \theta)$.

Many problems in maintenance fit such a framework. The classic age-based replacement model (Barlow & Proschan 1965) has $D = D(c; \alpha, \beta)$, where c is the critical age limit for replacement and α and β are the parameters of the hazard function. Extensions of this model (e.g. Boland & Proschan 1982; Sheu 1991) have $D = D(T; \theta)$, where θ parametrizes some more complex failure-repair model and $T, 2T, 3T, \dots$ are the times of preventive replacements. The overhaul model (Christer 1984) has $D = D(\Delta; \alpha, \beta)$, where Δ is the overhaul period and the cost of operation of the plant between overhauls is given by $C(t) = \alpha + \beta t$, for example. In practice, α and β require estimation from data. The condition-monitoring model due to Park (1988) also follows this general formulation, with $D = D(r; \lambda, \alpha)$ where r is the wear limit for preventive replacement and λ and α are the parameters of the gamma wear process. Models with multivariate decision variables are less common in the literature. Recently, for example, Christer & Wang (1994) have proposed a replacement model for a repairable system with multiple inspection intervals, for which $D = D(R, m, t_1, \dots, t_m; \theta)$. For the system when new, it is required to determine the time to replacement, R , the number m of inspections in $[0, R)$, and the times of these inspections, $t_1, \dots, t_m$.

Now denote the value of δ that corresponds to the minimum cost by δ^* , that is

$$\left[\frac{\partial D}{\partial \delta_i} \right]_{\delta = \delta^*} = 0 \quad (i = 1, \dots, p). \quad (2.1)$$

Let the corresponding minimum value of D be D^* . In general δ^* will be unknown and must be estimated. To do this, the nuisance parameters must be first estimated.

Denote the estimate of θ by $\hat{\theta}$. We optimize D as best we can in practice by solving the equations $\partial D(\delta; \hat{\theta})/\partial \delta_i = 0$ ($i = 1, \dots, p$), to obtain the estimated optimum decision variables $\hat{\delta}^*$. The subsequent inspection policy would then be operated with $\delta = \hat{\delta}^*$.

Now define the excess cost, ε , of operating at the estimated optimum $\hat{\delta}^*$ rather than the true unknown optimum δ^* by

$$D(\hat{\delta}^*; \theta) = D(\delta^*; \theta) + \varepsilon(\theta).$$

This excess cost arises since we are operating at $\hat{\delta}^*$ based on our estimates $\hat{\theta}$ of θ . Thus we wish to find the expectation of the excess cost, or risk, $E[\varepsilon(\theta)]$, in terms of the error $\hat{\theta} - \theta$. Expanding $D(\hat{\delta}^*; \theta)$ as a Taylor series in $\hat{\delta}^*$ about δ^* , it follows that

$$\varepsilon(\theta) = \sum_{i=1}^p (\delta_i^* - \hat{\delta}_i^*) \left[\frac{\partial D}{\partial \delta_i} \right]_{\delta=\delta^*} + \frac{1}{2} \sum_{i=1}^p \sum_{j=1}^p (\delta_i^* - \hat{\delta}_i^*)(\delta_j^* - \hat{\delta}_j^*) \left[\frac{\partial^2 D}{\partial \delta_i \partial \delta_j} \right]_{\delta=\delta^*} + \dots$$

Noting that the first term, from equation (2.1), is zero, we have as a first-order approximation

$$\varepsilon(\theta) = \frac{1}{2} \sum_{i=1}^p \sum_{j=1}^p (\delta_i^* - \hat{\delta}_i^*)(\delta_j^* - \hat{\delta}_j^*) \left[\frac{\partial^2 D}{\partial \delta_i \partial \delta_j} \right]_{\delta=\delta^*}. \quad (2.2)$$

We now need to express the error $\hat{\delta}^* - \delta^*$ in terms of the error $\hat{\theta} - \theta$. In general the following result may be used.

THEOREM 1 Let M be the $p \times p$ matrix with elements $m_{ij} = \partial^2 D / \partial \delta_i \partial \delta_j$ and T the $p \times q$ matrix with elements $t_{ij} = \partial^2 D / \partial \delta_i \partial \theta_j$, with all derivatives evaluated at (δ^*, θ) . Then, provided that M is positive definite,

$$\hat{\delta}^* - \delta^* \cong -M^{-1}T(\hat{\theta} - \theta). \quad (2.3)$$

COROLLARY $E[\varepsilon(\theta)] = \frac{1}{2} \text{tr}(T^T M^{-1}TV)$, where V is the matrix with elements

$$v_{ij} = E[(\hat{\theta}_i - \theta_i)(\hat{\theta}_j - \theta_j)]. \quad (2.4)$$

For the simplest models, it may be possible to express the optimum decision parameters δ^* as an explicit function of the nuisance parameters θ . In this case, the following simpler result may be used.

THEOREM 2 Let S be the $p \times q$ matrix with elements $s_{ij} = \partial \delta_i / \partial \theta_j$, with derivatives evaluated at $(\delta^*; \theta)$, then

$$\hat{\delta}^* - \delta^* \cong S(\hat{\theta} - \theta). \quad (2.5)$$

COROLLARY $E[\varepsilon(\theta)] = \frac{1}{2} \text{tr}(S^T M S V)$, where V is the matrix with elements

$$v_{ij} = E[(\hat{\theta}_i - \theta_i)(\hat{\theta}_j - \theta_j)],$$

and M is the positive definite $p \times p$ matrix with elements $m_{ij} = \partial^2 D / \partial \delta_i \partial \delta_j$.

Proof of these results is given in the Appendix. Here V is the covariance matrix of the nuisance-parameter estimates. The error in the estimates of the nuisance

parameters is given by $\sqrt{v_{ii}}$, where the v_{ii} are as in equation (2.4). If $V \propto 1/n$, where n is the sample size, then it follows immediately from the corollary to Theorem 1 that $E[\varepsilon(\theta)] \propto 1/n$. Since (provided that certain regularity conditions hold) maximum likelihood yields estimates with $V \propto 1/n$, this method of estimation will yield estimates of optimum decision variables with associated cost close to the true optimum value even for small sample sizes. This also implies that the error $\hat{\delta}_i^* - \delta_i^*$ in the decision-variable estimates is (from equations (2.3) and (2.5)) $\propto 1/\sqrt{n}$, and likewise for the nuisance-parameter estimates.

The matrix $T^T M^{-1} T$ may be determined numerically by solving the q sets of p equations $MB = T$ to determine B , and then $T^T M^{-1} T = T^T B$. In practice, to estimate the excess cost, we would need to replace M , V , and T (or S) by their estimates evaluated at $(\hat{\delta}^*, \hat{\theta})$.

3. A simple case for an exponential delay-time model

Consider the delay-time model for a repairable system proposed by Baker *et al.* (1993). In particular, for simplicity, we shall describe the above results in the context of the model for which: defects in the system arise according to a homogeneous Poisson process with rate α ; the times between defect arrival and resulting failure, the so-called delay times (Christer & Waller 1984a,b), are independently and exponentially distributed with mean $1/\gamma$; and, at inspection, all defects present are detected and repaired with probability 1. Assumptions concerning repair are considered fully in Baker *et al.* (1993), but we note here that repairs are neither minimal nor equivalent to renewals, but merely imply that the defect is removed from the system. We also consider the case in which inspections are carried out at a regularly spaced interval δ , with observation ceasing at the last inspection at time T . This case, although simple, however, is still of practical interest. The full likelihood for this model is

$$\ell = \left(\frac{1 - e^{-\gamma\delta}}{\gamma} \right)^{n-k} \left(\prod_{i=1}^k (1 - e^{-\gamma t'_i}) \right) \frac{(\alpha T)^n e^{-\alpha T}}{n!},$$

where $n - k$ defects are found at inspection and $t'_i < \delta$ ($i = 1, \dots, k$) are the times of failure since last inspection. Note that a sensible sample size here must include some failures, otherwise γ and α cannot be estimated.

In particular,

$$-\frac{\partial^2 \ln \ell}{\partial \gamma^2} = \sum_{j=1}^n \frac{t_j^2 e^{-\gamma t_j}}{(e^{-\gamma t_j} - 1)^2} - \frac{n-k}{\gamma^2},$$

where the summation is over all events (defects found at inspection and failures) and

$$t_j = \begin{cases} \delta & \text{if event } j \text{ is a detection of a defect at inspection,} \\ t'_i & \text{if event } j \text{ is failure } i. \end{cases}$$

Further $-\partial^2 \ln \ell / \partial \alpha^2 = -n - \omega/\alpha^2$, and $-\partial \ln \ell / \partial \gamma \partial \alpha = 0$. Thus the asymptotic covariance of (γ, α) is

$$V = \frac{1}{n} \begin{bmatrix} v_\gamma & 0 \\ 0 & \alpha^2 \end{bmatrix}, \quad (3.1)$$

where

$$v_\gamma = \left(\frac{\delta}{\gamma(e^{\gamma\delta} - 1)} + \frac{I(\gamma\delta)}{\gamma^3\delta} - \frac{1 - e^{-\gamma\delta}}{\gamma^3\delta} \right)^{-1}, \quad I(z) = \int_0^z \frac{x^2 dx}{e^x - 1}.$$

This is derived by noting that in general the asymptotic covariance matrix V of a parameter vector $\theta = (\theta_1, \dots, \theta_m)$ is defined by

$$V = \left\{ E - \frac{\partial^2 \ln \ell}{\partial \theta_i \partial \theta_j} : i, j = 1, \dots, m \right\}^{-1}.$$

Thus here $v_\gamma = n[E - (\partial^2 \ln \ell / \partial \gamma^2)]^{-1}$. The standard deviation of both $\hat{\gamma}$ and $\hat{\alpha}$ are $\propto \sqrt{n}$. An estimate of V is provided by replacing (γ, α) in equation (3.1) by their maximum-likelihood estimates. Note that it is also possible to estimate V using the observed information matrix

$$H^{-1} = - \left[\frac{\partial^2 \ln \ell}{\partial \theta_i \partial \theta_j} : i, j = 1, \dots, m \right]_{\theta = \hat{\theta}}^{-1}.$$

In many practical situations, this latter procedure is used since it is often difficult to find an expression for the asymptotic covariance. In fact, asymptotic arguments indicate that H^{-1} provides a better estimate of the covariance than the estimated asymptotic covariance (see Efron & Hinkley 1978). A useful method for calculating H^{-1} is described by Baker & Wang (1991).

We shall use the model proposed by Christer & Waller (1984b) for the cost per unit time, D , namely

$$D = \frac{c_1}{\delta} + \alpha c_2 [1 - P_{DC}(\gamma, \delta)],$$

where c_1 is the cost due to an inspection, c_2 is the cost due to a failure, and $1 - P_{DC}$ is the fraction of defects manifesting as failures. Here

$$P_{DC} = \frac{1 - e^{-\gamma\delta}}{\gamma\delta},$$

and so, the 1×1 matrix M is given by

$$m_{11} = \frac{\partial^2 D}{\partial \delta^2} = \frac{2c_1}{\delta^3} + c_2 \alpha \frac{e^{-\gamma\delta}(2 + 2\gamma\delta + \gamma^2\delta^2) - 2}{\gamma\delta^3}.$$

Now the condition $\partial D / \partial \delta = 0$ gives the equation for δ^* :

$$(1 + \gamma\delta^*) e^{-\gamma\delta^*} = 1 - \frac{\gamma c_1}{\alpha c_2}, \quad (3.2)$$

which has a solution, provided that $\gamma c_1 < \alpha c_2$. Thus, from equation (3.2),

$$s_{11} = \frac{\partial \delta}{\partial \gamma} = \frac{c_1 e^{\gamma \delta}}{\gamma^2 \delta c_2 \alpha} - \frac{\delta}{\gamma}, \quad s_{12} = \frac{\partial \delta}{\partial \alpha} = -\frac{c_1 e^{\gamma \delta}}{c_2 \gamma \delta \alpha^2},$$

where all derivatives are evaluated at $\delta = \delta^*$. It then follows that the expected excess cost of operating at the estimated optimum $\hat{\delta}^*$ rather than the true optimum δ^* is

$$E[\varepsilon(\gamma, \alpha)] = \frac{1}{2} \text{tr}(\mathbf{S}^T \mathbf{M} \mathbf{S} \mathbf{V}) = \frac{1}{2} m_{11} (s_{11}^2 v_\gamma + s_{12}^2 \alpha^2) / n. \quad (3.3)$$

An estimate of $E[\varepsilon(\gamma, \alpha)]$ is provided by replacing (γ, α) in equation (3.3) by their maximum-likelihood estimates. Note that $(s_{11}^2 v_\gamma + s_{12}^2 \alpha^2) / n$ also provides an estimate of the variance of $\hat{\delta}^*$; see Baker *et al.* (1993).

3.1 Numerical example

The maintenance department at a large hospital in Liverpool records the history of breakdowns and repairs for medical equipment. In particular, data on 2 identical ventilators used in the intensive-care unit were available for analysis. Over a period of 4741 machine-days, there had been a total of 16 failures and defects found at inspection. Over the period, inspections had been carried out approximately every 3 months. These data were analysed in Baker *et al.* (1993). The simple exponential delay-time model, with perfect inspection and arrival of defects according to a homogeneous Poisson process, was fitted to the data and gave estimates (and standard errors) $\hat{\gamma} = 0.0195$ (0.0087) and $\hat{\alpha} = 0.00337$ (0.00084). Thus the mean delay time was 51 days ($= 1/\hat{\gamma}$), and the defect arrival rate was approximately one every 300 days. The costs of inspection and failure were estimated at £45 and £370, respectively. The estimated cost, $\hat{D} = D(\delta; \hat{\gamma}, \hat{\alpha})$, is plotted against δ in Fig. 1. The estimated optimum inspection interval is $\hat{\delta}^* = 130$ days with a standard error of 45 days, and the estimated minimum cost is $\hat{D}^* = £416.4$ (per machine) per year. The estimated expected excess cost was £5.1 per year.

It is also of interest to consider the effect of increasing the sample size on the

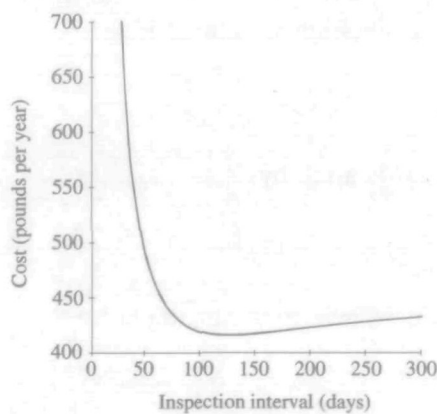


FIG. 1. Estimated cost per year, $\hat{D} = D(\delta; \hat{\gamma}, \hat{\alpha})$, versus length of the inspection interval, δ , for the ventilator data. ($\hat{\delta}^* = 130$ days approx. and $\hat{D}^* = £416.4$ per machine per year.)

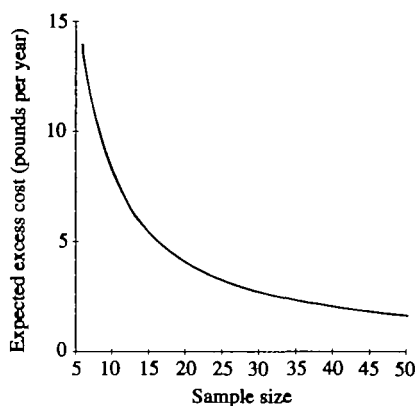


FIG. 2. Estimated expected excess cost, $\hat{E}[\varepsilon(\hat{\gamma}, \hat{\delta})]$, versus sample size n , with estimated optimum inspection interval $\hat{\delta}^* = 130$ days. (Ventilator data with $n = 16$ and $\hat{E}[\varepsilon(\hat{\gamma}, \hat{\delta})] = £5.1$ per machine per year).

estimated expected excess cost or risk. In Fig. 2, we have plotted $\hat{E}[\varepsilon(\hat{\gamma}, \hat{\delta})]$ against n , with c_1 and c_2 as above, and $\hat{\delta}^* = 130$ days. This plot illustrates the benefit or otherwise of collecting further data. If the cost of collecting further data were known, for example, then this cost could be compared with the reduction in expected excess cost as a result increasing the sample size n .

4. Discussion

Although large sample sizes are needed to estimate decision parameters, such as the optimum inspection interval, to high accuracy, it has been shown that the cost (or downtime) which is optimized can be reduced to near its minimum value even for small sample sizes, due to the quadratic nature of a minimum. In fact, the expected excess cost over the optimum (minimum) cost is inversely proportional to sample size. This means that even modest sample sizes would be sufficient for the practical use of sensible models for determining cost-effective maintenance policies. Thus for an OR consultant who wishes to optimize the inspection policy for a machine already running under a suboptimal policy, the crucial initial step is the calculation of the necessary sample size n to ensure an acceptable expected excess cost or risk, rather than an acceptable estimate of δ^* . If the management is able to specify an acceptable percentage of excess cost or downtime due to suboptimal running, then the necessary sample size would follow from this.

Further, this approach enables practitioners to quantify the benefit of collecting data in terms of the excess cost. If the cost of data collection (per sample point, say) is known, then it would be straightforward to optimize the sample size itself, given a set of training data on which to base parameter estimates. This could be done simply by comparing the cost of increasing sample size with the implied reduction in the expected excess cost of the maintenance policy.

It is often the case that more complex models can be reasonably supported (based on engineering judgement about the system) but that the extra parameters of these

models cannot be estimated with reasonable precision, given the sparsity of data relating to the system. It would seem sensible to choose the model that minimizes the expected excess cost. Minimizing the Akaike information criterion (AIC) (Sakamoto *et al.* 1986) can give the 'best' model asymptotically, and hence roughly speaking minimize the expected excess cost. This is not guaranteed, however, and we refer the reader to Choi (1992) for more stringent criteria. However, all of this implies that the practitioner needs at the same time to adopt some rational method for model selection, such as the AIC, for the results in this paper to be most useful.

Acknowledgements

The authors would like to express their gratitude to the medical-equipment staff at Liverpool Royal Hospital, in particular to Dr. Malcolm Brown, Head of Bio-engineering, for supplying the data, and to our colleague Dr. Wenbin Wang for help with the analysis. This work has been supported in part by a grant from the SERC, grant no. GR/H8/1535.

Appendix

Proof of Theorem 1. Expand $D(\hat{\delta}^*; \hat{\theta})$ in its Taylor series about $(\delta^*; \theta)$. Thus

$$\begin{aligned} D(\hat{\delta}^*; \hat{\theta}) &= D(\delta^*; \theta) + \sum_{i=1}^p (\hat{\delta}_i^* - \delta_i^*) \frac{\partial D}{\partial \delta_i} + \sum_{k=1}^q (\hat{\theta}_k - \theta_k) \frac{\partial D}{\partial \theta_k} \\ &\quad + \frac{1}{2} \sum_{i=1}^p \sum_{j=1}^p (\hat{\delta}_i^* - \delta_i^*)(\hat{\delta}_j^* - \delta_j^*) \frac{\partial^2 D}{\partial \delta_i \partial \delta_j} \\ &\quad + \frac{1}{2} \sum_{k=1}^q \sum_{l=1}^q (\hat{\theta}_k - \theta_k)(\hat{\theta}_l - \theta_l) \frac{\partial^2 D}{\partial \theta_k \partial \theta_l} \\ &\quad + \sum_{i=1}^p \sum_{k=1}^q (\hat{\delta}_i^* - \delta_i^*)(\hat{\theta}_k - \theta_k) \frac{\partial^2 D}{\partial \delta_i \partial \theta_k} + \dots, \end{aligned}$$

where all derivatives are taken at $(\delta^*; \theta)$. Now differentiating both sides with respect to $\hat{\delta}_i^*$ ($i = 1, \dots, p$), and noting that $\partial D(\hat{\delta}^*, \hat{\theta}) / \partial \hat{\delta}_i^* = 0$, we get (recalling that $\partial D / \partial \delta_i = 0$, and that $D(\delta^*, \theta)$, $\partial D / \partial \theta_k$, $\partial^2 D / \partial \delta_i \partial \delta_j$, etc. are all constant with respect to $\hat{\delta}_i^*$)

$$0 = \sum_{j=1}^p (\hat{\delta}_j^* - \delta_j^*) \frac{\partial^2 D}{\partial \delta_i \partial \delta_j} + \sum_{k=1}^q (\hat{\theta}_k - \theta_k) \frac{\partial^2 D}{\partial \delta_i \partial \theta_k} + \dots \quad (i = 1, \dots, p).$$

Hence $M(\hat{\delta}^* - \delta^*) \cong -T(\hat{\theta} - \theta)$, where M is the $p \times p$ matrix with elements $m_{ij} = \partial^2 D / \partial \delta_i \partial \delta_j$, and T is the $p \times q$ matrix with elements $t_{ij} = \partial^2 D / \partial \delta_i \partial \theta_j$. This completes the proof of Theorem 1 on noting that M is invertible, since it is a symmetric positive definite matrix.

Proof of the Corollary to Theorem 1. This follows on noting that, from equation (2.2),

$$\varepsilon(\theta) = \frac{1}{2}(\hat{\delta}^* - \delta^*)^T M(\hat{\delta}^* - \delta^*) = \frac{1}{2}(\hat{\theta} - \theta)^T T^T M^{-1} T(\hat{\theta} - \theta) = \frac{1}{2} \text{tr}(T^T M^{-1} T W),$$

where the elements of W are $w_{ij} = (\hat{\theta}_i - \theta_i)(\hat{\theta}_j - \theta_j)$. Thus

$$E[\varepsilon(\theta)] = \text{tr}(T^T M^{-1} T V),$$

where $V = E[W]$.

Proof of Theorem 2. Expanding $\hat{\delta}^*(\hat{\theta})$ in its Taylor series about θ , we get

$$\hat{\delta}_i^* = \delta_i^* + \sum_{j=1}^q (\hat{\theta}_j - \theta_j) \frac{\partial \delta_i^*}{\partial \theta_j} + \dots \quad (i = 1, \dots, p),$$

where all derivatives are taken at (δ^*, θ) . Thus $(\hat{\delta}^* - \delta^*) \cong S(\hat{\theta} - \theta)$, where S is the $p \times q$ matrix with elements $s_{ij} = \partial \delta_i / \partial \theta_j$.

Proof of the Corollary to Theorem 2. This follows similarly to the proof of the Corollary to Theorem 1 above. Note that this result and the Corollary to Theorem 1 imply that $S = -M^{-1}T$. This result may be illustrated for the case $p = q = 1$. Since by definition $\partial D / \partial \delta = 0$ at δ^* , considering the total differential we get,

$$\frac{\partial^2 D}{\partial \delta^2} d\delta + \frac{\partial^2 D}{\partial \delta \partial \theta} d\theta = 0,$$

and so

$$\frac{\partial^2 D}{\partial \delta^2} \frac{\partial \delta}{\partial \theta} + \frac{\partial^2 D}{\partial \delta \partial \theta} = 0.$$

Thus

$$\frac{\partial \delta}{\partial \theta} = - \left(\frac{\partial^2 D}{\partial \delta^2} \right)^{-1} \frac{\partial^2 D}{\partial \delta \partial \theta},$$

where the derivatives are taken at $\delta = \delta^*$.

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